

PushT Follow-Up Addendum

AF-LeWM-lite v2 Sweep and Stability Check

April 10, 2026

Updated conclusion. The original AF-LeWM-lite v2 objective still gives the best 50-episode PushT control result, but the advantage over baseline is small and unstable. A second 50-episode evaluation seed reduces the gap to **5/100** successes for v2 versus **4/100** for baseline. The current evidence supports the existence of a real representation-control tradeoff, and it does not yet support a strong robustness win.

Why this follow-up

The first official PushT report showed a tension:

- baseline had the best shared JEPA objective,
- original v2 had the best 50-episode success,
- the gap was only one success: 3/50 versus 2/50.

This addendum tests whether the adversarial nuisance term is responsible for both the control gain and the optimization damage.

Fair comparison metric

Cross-model loss comparison uses the shared objective

$$\mathcal{L}_{\text{core}} = \mathcal{L}_{\text{pred}} + 0.09 \mathcal{L}_{\text{sigreg}},$$

because total training loss includes AF-specific auxiliary terms and is not directly comparable once those terms are active.

Sweep results

Model	Core loss	Success (%)	Comment
Baseline LeWM	0.1469	4.0	reference
AF-LeWM-lite v2	0.4726	6.0	original best control
v2 no adversary	0.1491	0.0	removed dynamics nuisance adversary
v2 soft	0.3603	2.0	reduced all auxiliary weights
v2 grl half	0.3145	2.0	halved GRL only

What changed

Removing the adversary restores the shared world-model loss and destroys control. The no-adversary run lands almost on top of baseline in core loss, then falls to 0/50 successes. That is the strongest sign that the dynamics-side nuisance adversary is doing real work for control.

Weakening the adversary improves the loss and weakens control. Both the soft variant and the GRL-half variant move the core loss back toward baseline, and both drop to 1/50 successes.

The current tradeoff is sharp. Better shared modeling and better PushT control are pulling in different directions under this short budget.

100-episode check

I reran baseline and the original v2 on a second evaluation seed while keeping the same 50-episode protocol per seed.

Model	Seed 42	Seed 43	Aggregate
Baseline LeWM	2/50	2/50	4/100 = 4.0%
AF-LeWM-lite v2	3/50	2/50	5/100 = 5.0%

This keeps v2 slightly ahead, and the margin is too small to claim a robust improvement.

Research conclusion

- The dynamics nuisance adversary is the key mechanism behind the current v2 control gain.
- The same mechanism is also the main source of degradation in the shared JEPA objective.
- The present PushT evidence supports a **tradeoff**, not a clear overall win.
- The next experiment should tune **when** the adversary turns on, not only **how strong** it is. A delayed or ramped adversary schedule is the highest-value next test.

In one sentence: **the factorized-latent idea still looks alive, but the current v2 benefit is fragile and the right next move is schedule tuning rather than static weight reduction.**